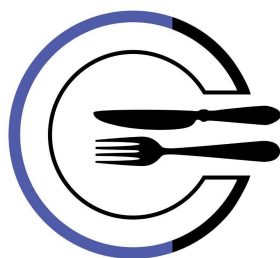


CALORIETOKEN

From a food-token idea to a verifiable food-first product

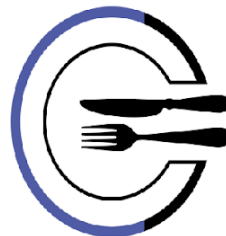
Current CalorieApp, historical development and future research, with their status clearly explained.



Version 4.2 | 10 September 2026

ICTHendrikse | KVK 73774693
Project maintained by Pieter Hendrikse

This edition updates website integrations, food-source information and community status. Local and repository checks do not establish that the latest package is installed or that every live device journey has been accepted.



00 How to read this revision

The strongest public claim is the one whose status can be checked.

CURRENT WEBAPP

Implemented in the current public CalorieApp preview and described within a bounded, non-financial product scope.

HISTORICAL

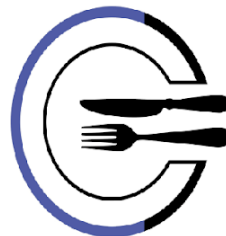
Earlier prototypes, testnet experiments and published concepts. Useful context, but not the current product architecture.

FUTURE / RESEARCH

A direction or research area only. It is not a live feature, delivery promise, token utility commitment or announced launch schedule.

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01 Executive summary

CalorieToken now has a simpler public center: an observable food and nutrition workflow.

CalorieToken began in 2021 with a broad food, payments and tokenisation vision. Several prototypes explored that vision. The current public product is deliberately narrower: CalorieApp is a non-financial, non-custodial web application for searching food data, reviewing nutrition information and maintaining a personal food log.

WHAT EXISTS NOW

Public food search, nutrition result display, portion-aware logging, authenticated retrieval and user-scoped deletion. Xaman is used only as an external identity boundary.

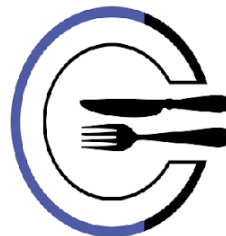
WHAT DOES NOT EXIST IN THE CURRENT WEBAPP

No custody, private-key handling, transaction signing, payments, token transfers, trading, investment services or on-chain food-log storage.

WHY THIS REVISION MATTERS

Version 3.1C mixed a future ecosystem concept with legacy testnet implementation. Version 4.2 preserves that history while aligning current product claims with the public repository and hosted webapp.

The future direction remains ambitious - food data quality, privacy-conscious research, traceability and carefully gated interfaces - but every future capability must earn its status through evidence, review and public documentation.



02 Mission and design principles

Make food information more usable, then grow only from verified foundations.

MISSION

Build accessible food and nutrition workflows that people can inspect, use and improve, while keeping financial and identity boundaries explicit.

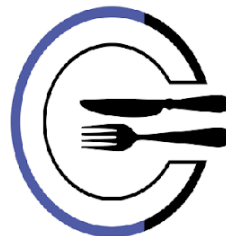
The original vision connected food, community and digital infrastructure. The current strategy keeps those themes but changes the order of operations: prove the food workflow first; evaluate broader ecosystem layers later.

Design principles

- Public proof before public promises. Current claims should be demonstrable in a repository, hosted product or public record.
- Food-first scope. The product starts with nutrition search and personal logging, not with financial execution.
- User control. Authenticated users can retrieve and delete their own food-log entries.
- External identity boundary. Xaman remains outside the application; CalorieApp does not receive custody or signing authority.
- Evidence-gated expansion. Research themes become product claims only after technical, privacy, legal and operational review.
- Clear publication status. Historical material and future concepts remain visible, but never masquerade as current webapp.

NORTH STAR

A credible food ecosystem is built by making each layer testable: user experience, data quality, privacy, interfaces, governance and - only if separately approved - any future token-related utility.



03 Current public status

A bounded preview with a real end-to-end food workflow.

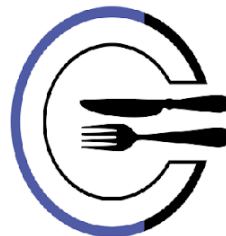
PRODUCT	SCOPE	IDENTITY
CalorieApp	Food & nutrition	External Xaman
Public preview	Non-financial	Non-custodial

Implemented in the current webapp

- Search public product records through Open Food Facts.
- Display nutrition information supplied by the external data source.
- Choose a portion percentage and add an item to a personal food log.
- Retrieve the authenticated user's stored log entries.
- Delete entries within the authenticated user's own scope.
- Expose a frontend/backend health path and session-based application flow.

PUBLIC EVIDENCE	CURRENT LIMIT
Open repository, public documentation, hosted preview and visible release checks.	A preview can demonstrate the workflow without claiming medical accuracy, full production readiness or financial functionality.

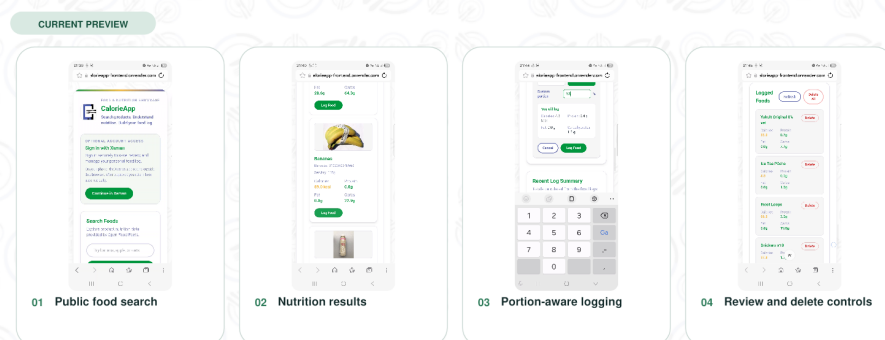
September 2026 update: the repository includes a small USDA reference-food selection, eleven-language interface controls, diary filtering and clearer startup/rate-limit feedback. The latest WordPress package requires installation and a live acceptance check; the user has not yet accepted the updated app journey.



04 CalorieApp food journey

Search, understand, log and control personal entries on a real phone-sized interface.

Today's preview shows a food journey on a real phone



Current public preview: food search, nutrition results, portion-aware logging, and review/delete controls.

PUBLIC SEARCH

Food search is available without turning the application into a wallet or payment interface.

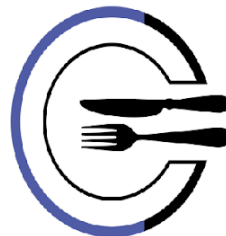
PERSONAL LOG

Saving, reviewing and deleting entries requires the application's identity and session boundary.

WHAT THIS PROVES

The project now has an observable food workflow. It does not by itself prove clinical suitability, complete product-data accuracy, scale under all loads or readiness for regulated financial functions.

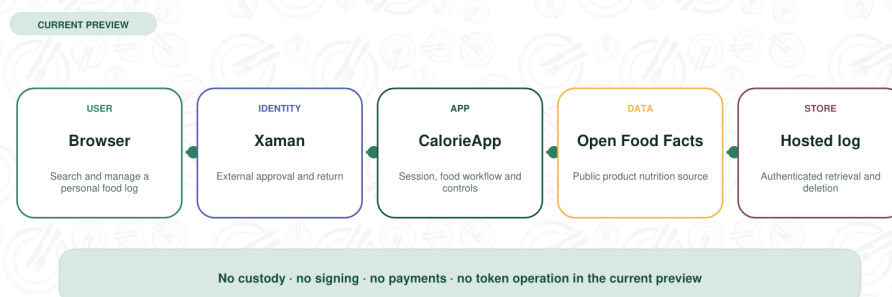
A slow start keeps the search query in place. Repeated clicks do not restart the pending search. A provider-directed pause is shown before a manual retry; availability still depends on the host and food provider.



05 Current architecture

Identity, application logic, public food data and personal logging remain separate concerns.

Authentication is an external boundary—not a wallet surface



Website and CalorieApp sessions connect through one Xaman sign-in; personal food logs remain in the application.

The frontend is built with Next.js, TypeScript and Tailwind CSS. The backend uses FastAPI and SQLAlchemy with an application database for current webapp. Open Food Facts supplies public product and nutrition data. Xaman is an external identity context, not a wallet surface inside CalorieApp.

SESSION BOUNDARY

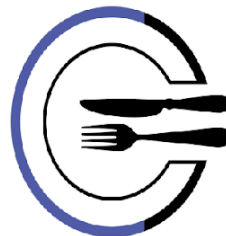
One Xaman sign-in connects the website and CalorieApp in the same browser. Combined logout ends both sessions.

DATA BOUNDARY

Personal log operations are user-scoped. Public product records remain distinct from personal entries.

NOT IN THIS ARCHITECTURE

No private keys, seed phrases or signing credentials are requested or stored. No payment, token-transfer, trading, validator or ledger-node operation is performed by the webapp.



06 Privacy, security and regulatory boundary

The public scope is intentionally non-financial and non-custodial.

PRIVACY

Minimise personal data; separate public product data from personal food logs; keep retrieval and deletion user-scoped; document future data uses before implementation.

SECURITY

Treat external identity as a boundary; keep secrets out of public code and documents; test release paths; preserve rollback and recovery evidence.

Current webapp explicitly excludes

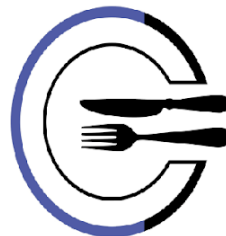
- Custody of assets or credentials.
- Transaction signing, token transfers, payments or trading.
- Investment services, financial advice or yield/reward promises.
- Medical diagnosis, personalised treatment or clinically validated nutrition advice.
- Public exposure of private food logs or sensitive operational evidence.

CHANGE GATE

Any future financial, token, reward, profiling or health-related feature requires a separate architecture, security, privacy and legal review before it can be described as a product capability.

This document records a product boundary; it is not legal, financial or medical advice and does not claim regulatory certification.

Website-only integrations remain separate from the food app: external CAL/XRP routes, SWFT, Testnet guidance and the local help guide do not turn CalorieApp into a trading or custody application. Private import, whole-account erasure and pilot expansion retain their individual release gates.



07 Food data and nutrition context

Useful product information depends on source quality, context and user judgement.

CalorieApp queries Open Food Facts, a collaborative public food-products database. The source makes a broad catalogue available, but records can be incomplete, inconsistent, outdated or supplied in different serving conventions.

SOURCE

Product identity, ingredients and nutrition fields are provided by Open Food Facts when available.

APPLICATION

CalorieApp uses complete source values per serving, or a labelled 100 g/ml reference. Portion percentages apply to that amount.

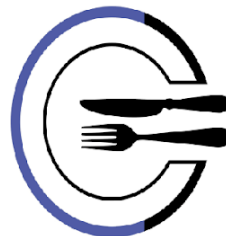
Responsible use

- Check the physical product label when accuracy matters.
- Check the displayed serving or reference amount before choosing a portion percentage.
- Avoid presenting aggregate observations as personal health conclusions.
- Record source attribution and applicable reuse obligations in release documentation.
- Separate future research datasets from current personal food-log processing.

DATA QUALITY ROADMAP

Before broader analysis or traceability claims, define provenance, update frequency, validation rules, correction flows and permission boundaries. Quality controls are a prerequisite, not a later cosmetic layer.

The current repository also contains three dated USDA FoodData Central reference foods with source links and a stated 100 g basis. This is a small bundled reference selection, not live USDA search. USDA reference data and Open Food Facts records retain separate attribution and reuse conditions.



09 Historical development

Earlier testnet work remains part of the record, but not part of the current webapp architecture.

The 2026 v3.1C edition appended a development update covering a legacy Kivy/KivyMD testnet application. That prototype explored local XRPL wallet creation/import, multiple accounts, issued-token balances, transaction history, endpoint failover, food-data entry, DEX/NFT screens and CalorieDB concepts.

WHAT WAS LEARNED

The work exposed integration complexity, recovery requirements, mobile UX constraints and the need to separate on-chain and off-chain responsibilities.

WHAT CHANGED

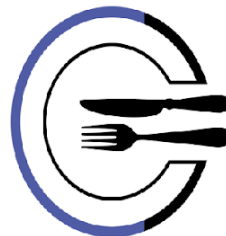
The current Next.js/FastAPI webapp removed wallet custody and financial execution from its product surface, narrowing the public proof to food and nutrition.

Historical interpretation rules

- A prototype proves exploration, not production readiness.
- A screen or subsystem in an archive is not a current product feature.
- Legacy private-key handling must not be inferred as part of present CalorieApp.
- Unfinished DEX, NFT, payment and database concepts remain historical or future research.
- Public history is preserved to explain the change in direction, not to inflate the current scope.

WHY PRESERVE THE HISTORY

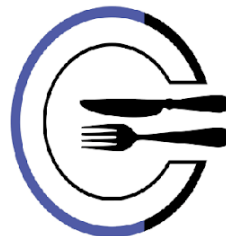
The development arc is useful when it is dated and labelled. It shows which assumptions were tested, which boundaries changed and why today's narrower product is the more credible public foundation.



10 Future and research direction

Direction is preserved, but every item remains conditional until evidenced.

CalorieDB A privacy-reviewed research layer for structured food data and provenance.	Traceability Optional supply-chain and integrity workflows built around verifiable references.
Content addressing Evaluation of IPFS/Helia or comparable systems for public artefact integrity.	Ledger references Potential hash correlation without publishing personal food-log content on-chain.
Native experiences Mobile applications and improved recovery/accessibility after the web flow stabilises.	Community infrastructure Clear contribution paths, documented proposals and bounded participation experiments.
NOT A DELIVERY COMMITMENT These directions have no guaranteed schedule, token utility, partner adoption, funding outcome or production status. Publication here records research intent only.	
PROMOTION RULE A future capability moves into public product marketing only after a working implementation, documented boundaries, reproducible evidence and the required privacy, security, legal and operational review.	



11 Evidence-gated roadmap

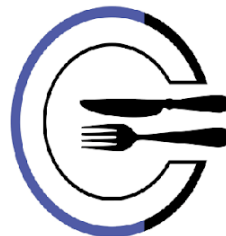
Progress is defined by evidence gates instead of speculative dates.

GATE 1	Stable webapp	Food journey, identity return, accessibility, quality and recovery evidence.
GATE 2	Pilot data	Partner samples, source provenance and measurable data-quality rules.
GATE 3	Interfaces	Documented APIs, data contracts, error handling and permission boundaries.
GATE 4	CalorieDB research	Privacy-reviewed storage model, retention logic and controlled evaluation.
GATE 5	Traceability pilots	Only after governance, legal review, partner evidence and reversible pilots.

GATE RULE

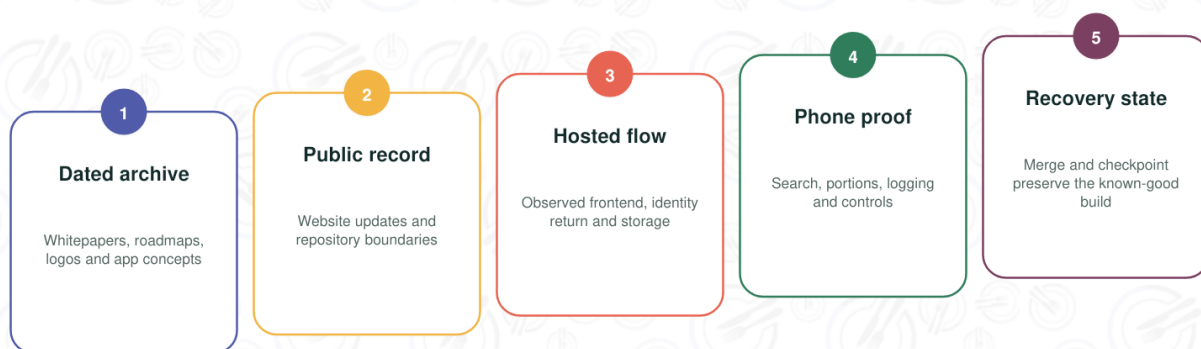
Each gate must be evidenced before it changes from direction to product claim. Gates can pause, repeat or close; no gate is a promise of release.

- Publish the evidence expected for a gate before announcing completion.
- Treat privacy, security and legal findings as release conditions, not footnotes.
- Prefer reversible pilots and measured learning over irreversible commitments.
- Keep sensitive findings in the internal evidence layer; publish only safe conclusions.



12 Engineering and release discipline

Public confidence grows from repeatable checks, not feature count alone.



Evidence ladder used for public claims: archive, public record, hosted flow, phone proof and recoverable state.

QUALITY GATES

Backend tests, frontend lint/build, repository-boundary checks and plugin/release checks protect the current code path.

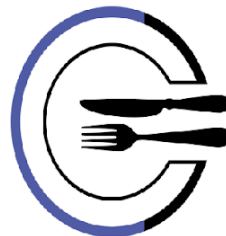
RECOVERY

Known-good states, branches, checkpoints and rollback notes reduce the cost of a failed change.

Release readiness also includes secret scanning, licence and provenance checks, privacy review, documentation accuracy and a confirmation that public claims match the actual release.

PUBLIC-SAFE EVIDENCE

Publish outcomes and reproducible public artefacts. Keep credentials, private logs, exploit detail, sensitive architecture and operational response material in restricted internal evidence only.



13 Governance and publication discipline

Participation works when decisions are bounded, understandable and verifiable.

CalorieToken has experimented with community participation. Version 4.2 does not present those experiments as decentralised control over funds or as a mature DAO. The public lesson is narrower: participation needs a simple question, clear scope, understandable consequences and a documented outcome.

PUBLIC LAYER

Verified product facts, safe architecture summaries, reproducible links, dated decisions and clearly labelled research directions.

INTERNAL LAYER

Sensitive findings, private or confidential evidence, credentials, incident detail, operational weaknesses and unresolved security material.

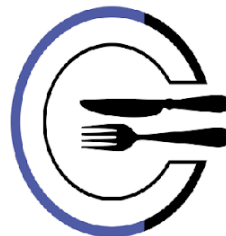
A proposal is ready for public participation when

- The question can be understood without specialist knowledge.
- The decision rights and implementation authority are stated honestly.
- The choices are limited enough to evaluate.
- Privacy, legal and financial implications have been reviewed.
- The result and follow-up can be recorded in a public decision log.

MARKETING GATE

Information is not published merely because it exists. Public content must be verified, audience-relevant and strategically helpful. Weak, confusing, private, sensitive or confidential material belongs in the internal layer or remains on hold.

The informational Community Voting Hub is prepared for public reading with historical 2024 illustrations. No vote, proposal submission or voting function is active. No current reward allocation or entitlement follows from those illustrations. Publication of the new page occurs when the WordPress update is installed.



14 Risks, limitations and disclaimers

This whitepaper describes status and direction; it does not remove uncertainty.

Product maturity

The public webapp is a bounded preview under active QA. Availability, latency and behaviour may change.

External dependencies

Identity, hosting and food data depend on external providers. Startup and searches may be delayed or temporarily unavailable.

Data quality

Public product records may be incomplete, inconsistent or outdated. Check physical labels when accuracy matters.

Security and privacy

No system is risk-free. Scope expansion may introduce new threat, consent and retention requirements.

Regulation

Future token, payment, reward, profiling or health functions may be restricted, redesigned or abandoned after review.

Execution

Research directions depend on evidence, capability, partners, funding and governance; they may not ship.

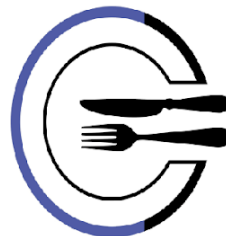
NO FINANCIAL OFFER

Nothing in this document is an offer, solicitation, investment recommendation, return promise or statement that CAL has a particular current or future value. Digital assets can be volatile and may lose all value.

NO MEDICAL ADVICE

Nutrition information is general product data, not diagnosis, treatment or personalised medical advice. Consult qualified professionals for health decisions.

Use of CalorieApp, external services and CAL remains subject to applicable terms, privacy notices and law. Independent professional advice may be required.



15 Sources and version history

A stable public document should show where its claims came from and what changed.

Primary public sources

- [Current CalorieApp repository](#)
- [Public architecture](#)
- [Public roadmap](#)
- [Identity boundary](#)
- [Regulatory boundary](#)
- [Latest prior whitepaper v3.1C](#)
- [CAL public explorer listing](#)
- [CAL issuer account on Bithomp](#)
- [Open Food Facts](#)

Version history

v4.1	8 Sep 2026	Updated web integration, portion basis and availability notes; original wave artwork retained.
v3.1C	19 Feb 2026	Legacy concept plus appended testnet progress update; superseded for current product claims.
v3.1	09 Apr 2023	Original long-form ecosystem concept and visual design reference.

CANONICAL PUBLICATION

The stable GitHub path is `whitepaper/CalorieToken-Whitepaper.pdf`. Dated PDFs remain in the public release archive. Editable working sources are retained separately; earlier editions are not overwritten.

v4.2 - 10 Sep 2026: original full-resolution historical background restored; selectable text and clearer tables; USDA, external website services, inactive voting and current acceptance limits documented. v4.1 remains in the immutable 8 September archive.

[USDA FoodData Central](#)

[USDA data documentation](#)

[MiCA: ESMA Q&A 2671](#)